

Open Source Smart Espresso Machine

TED Talk, 4/3/26

Students: Hussein Ashkanani, Ethan Budde, David Mazhnikov, Roman Sulza, Clarke Ren

Industry Sponsor: Dr. Elliott Gall

Faculty Advisor: Dr. Joshua Méndez



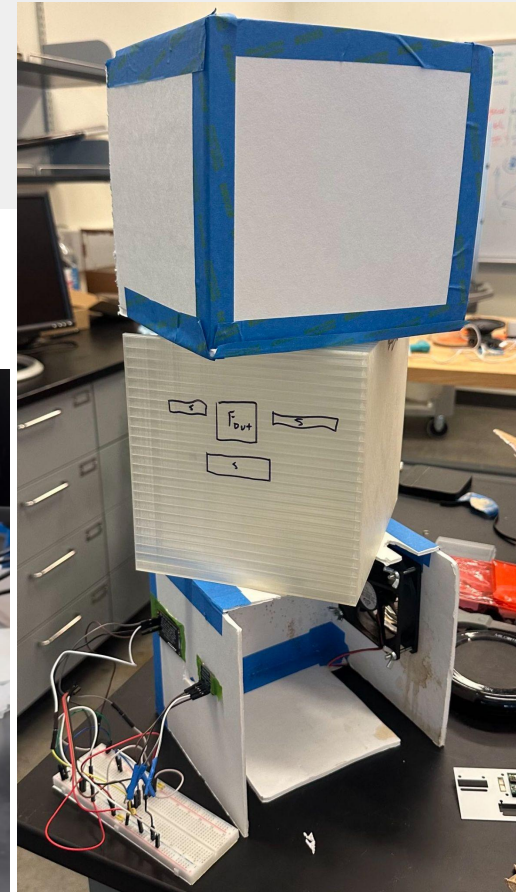
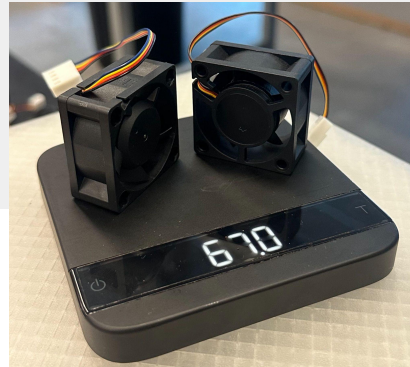
Who are we again?

- PSU **OsSEM**, Open-source Smart Espresso Machine
- Sponsored by **Healthy Buildings Research** and **Coffee & Telesensing Laboratories**
- Functional enhancement modification for Gaggia Classic espresso machine
 - powered by STM32 and ESP32 hardware
 - open-source software/firmware installations



Updates:

- ECE:
 - Bluetooth scale integrated
 - Data telemetry (mostly) operational
 - Soft redesign of fan controller board
 - Local network for data transfer being built (slowly)
- ME:
 - 3D printed enclosure(s) for shape testing, fan/sensor position variance
 - CNC machining for enclosure rev 3
 - Machined bottomless Portafilter



Updates:

- What's going great:
 - We can make coffee that doesn't (entirely) suck
- What's going not-so-great:
 - How do you sleep when your bloodstream is caffeine?
- What we wish we knew:
 - The location of a secret cache of riches (gold doubloons, gems, etc)
- Surprises?
 - ...

